(0) Key layer

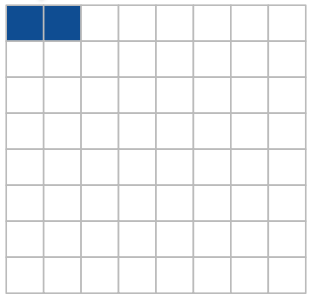
K, u

HKDF

k_u

coord PRF E_u(i,j)

payload PRF M(i,j)



E_u(i,j) = PRF(k_u, i, j)

(a) Publish-time embedding



I

QIM embedding
8×8 DCT, δ = 15
(1,0) = M, (1,1) = E_u



I_w

distribution channel (editing · JPEG · feathering)

(b) Blind auditing



I′

block verify vs E_u → F



F

detect top-10% / contrast stat > 0.41 ?

yes

EDITED

region alarm 0.38 + CC

4×4 refine (gated)

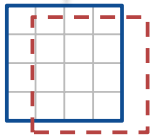
no

BENIGN

sync 64 phases × offsets low10 < 0.20 ?

yes

ABSTAIN degenerate



offset search